

Investment and Asset Prices

Macroeconomics B
Lecture 1 on Chapter 15

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Roadmap

- ▶ Why investment matters for business cycles
- ▶ Stylized facts: asset prices and the cycle
- ▶ Bonds, shares, discounting, and present value
- ▶ Why stock prices move a lot
- ▶ Tobin's q as the bridge to investment
- ▶ Short active exercise with FRED and Python

What You Should Know After This Lecture

- ▶ Read basic empirical figures on investment and asset prices
- ▶ Explain present value and discounting
- ▶ Derive the one-period stock pricing equation step by step
- ▶ Explain how r , risk premia, and expected profits affect prices
- ▶ Define Tobin's q and explain why it matters for investment

Why Investment Matters in Macroeconomics

- ▶ In the short run, output is demand determined:

$$Y = C + I + G.$$

- ▶ Investment is the most volatile component of demand.
- ▶ Because of that, investment is central for booms and recessions.
- ▶ To understand investment, we need to understand asset prices.

Types of Investment

- ▶ **Business fixed investment:** machines, equipment, software, structures.
- ▶ **Housing investment:** residential construction.
- ▶ **Inventory investment:** changes in stocks of goods.

This lecture focuses on the asset-pricing logic behind business and housing investment.

The Core Idea of Chapter 15

- ▶ The key question is: *Is one more unit of capital worth more than it costs to buy and install?*
- ▶ If yes, firms want to invest more.
- ▶ For housing, the same logic applies: if a house sells for more than it costs to build, construction rises.
- ▶ Economists summarize this with **Tobin's q** :

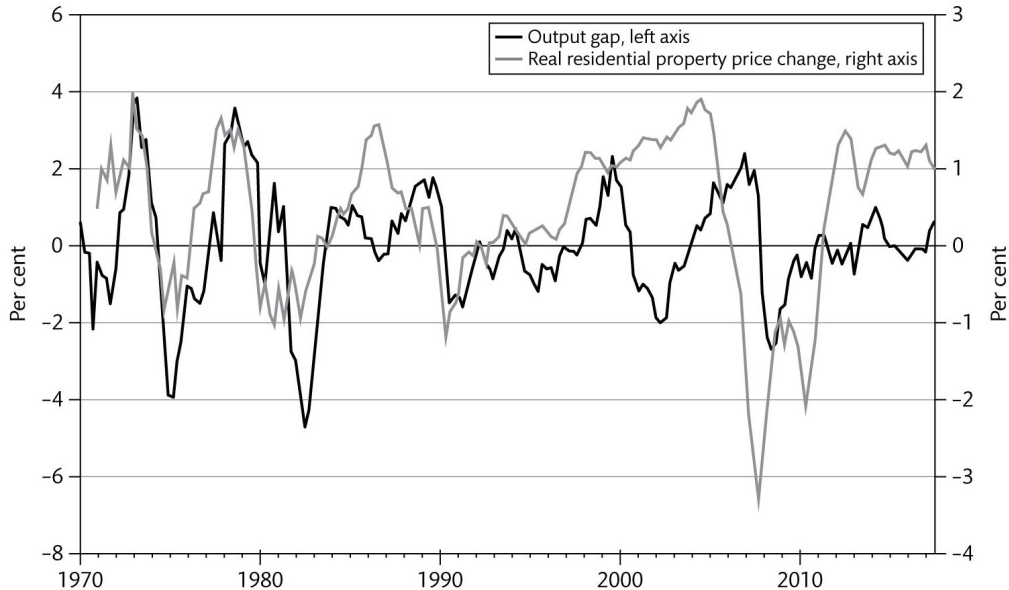
$$q = \frac{\text{value of installed capital}}{\text{replacement cost}}$$

- ▶ If $q > 1$, investment is attractive. If $q < 1$, investment is unattractive.

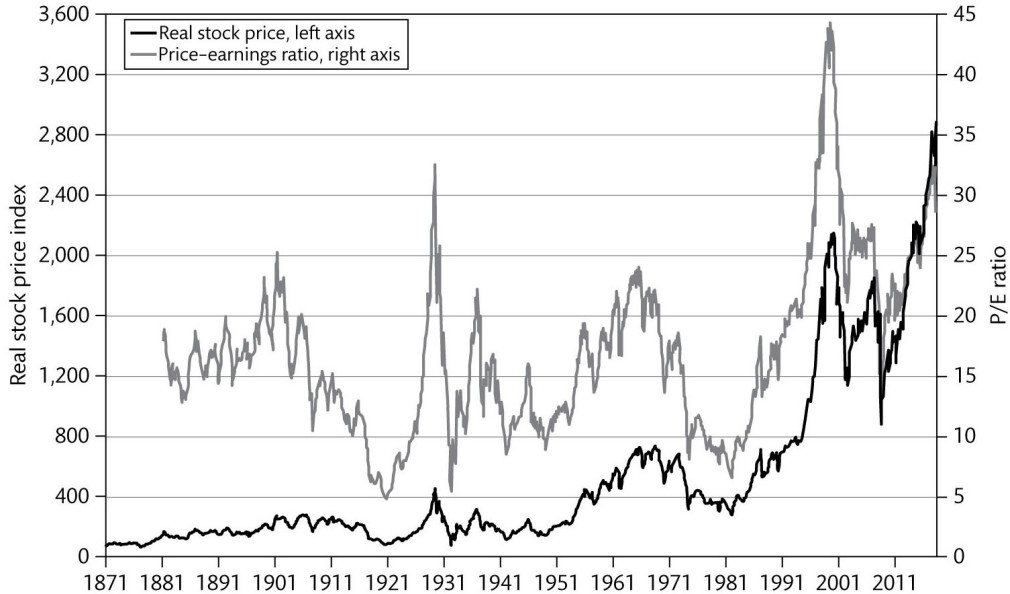
Example 1: Firm. A machine costs 1 million kr. to install, but financial markets value the extra capital at 1.2 million kr. Then $q = 1.2$, so investment is profitable.

Example 2: Housing. A new flat can be sold for 3.6 million kr., while building it costs 3 million kr. Then $q = 1.2$, so more construction is attractive.

Stylized Fact 1: Property Prices and the Cycle



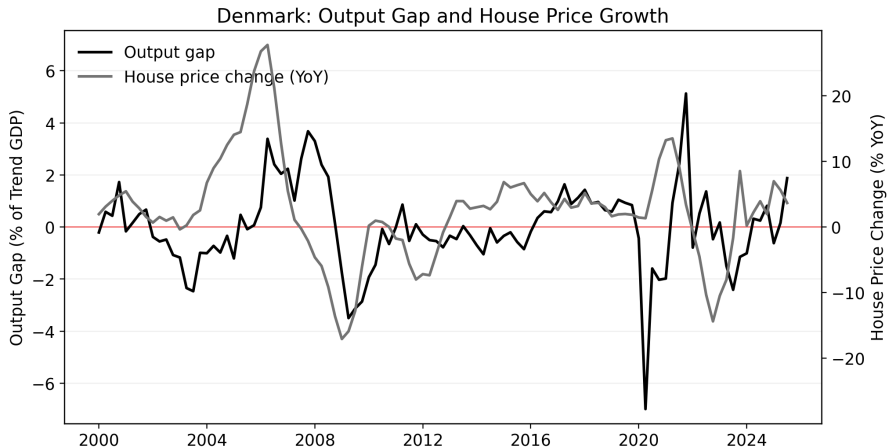
Stylized Fact 2: Stock Prices and Valuation over the Cycle



What the Two Figures Tell Us

- ▶ Both equity prices and house prices are strongly procyclical: they rise in booms and weaken in downturns.
- ▶ Asset prices are forward-looking, so they often move before GDP and investment do.
- ▶ When stock prices jump, firms usually read that as a signal that future profits look stronger.
- ▶ When house prices rise, construction and housing activity often pick up with a lag.

Denmark: Output Gap and House Price Growth



Danish house price growth is procyclical and often turns early.

Sources: Eurostat via FRED (Real GDP) and BIS via FRED (Residential Property Prices for Denmark). The Output Gap Is Constructed as an HP-Filtered Log Real GDP Series. Sample Through 2025Q3.

Think-Pair-Share: Asset prices as leading indicators

Think (1 min): Write down one reason why stock prices might fall *before* investment drops.

Pair (2 min): Compare with your neighbour. Do you agree?

Share: We'll collect 2–3 answers.

Hint: think about expectations, discount rates, and credit.

The price of stock

Objective of the firm

- ▶ Firms maximize the *market value* of outstanding shares
- ▶ Equivalent to maximizing owners' wealth and consumption possibilities

Arbitrage condition

- ▶ Share prices adjust until stocks and risk-free bonds offer equal expected returns
- ▶ No arbitrage profit is possible in equilibrium

Fundamental value

- ▶ Market value = discounted value of expected future cash flows (dividends)

Profit vs. market value

- ▶ No contradiction: maximizing the discounted profit stream also maximizes market value
- ▶ Profits are the source of cash flows to owners

The Gordon Growth Model: Notation and Timing

- ▶ V_t : stock price at the **beginning** of period t
- ▶ V_{t+1}^e : expected stock price at the beginning of period $t + 1$
- ▶ D_t^e : expected dividend paid at the **end** of period t
- ▶ r : real interest rate on bonds
- ▶ ε : equity risk premium

If an investor buys a share at time t , the expected payoff over the period is

$$D_t^e + V_{t+1}^e.$$

Bonds: The Simplest Asset

A one-period bond costs P_t^B today and pays \$1 with certainty next period $(1 + r) P_t^B = 1$.

Price (what you pay)

$$P_t^B = \frac{1}{1 + r}$$

Determined by no-arbitrage.

Yield (what you earn)

$$r = \frac{1}{P_t^B} - 1$$

Return from holding to maturity.

The inverse relationship

▶ $r \uparrow \Rightarrow P_t^B \downarrow$

▶ $r \downarrow \Rightarrow P_t^B \uparrow$

▶ Price and yield always move in opposite directions

The bond price is the discount factor. The same present-value logic will price stocks - only the payoffs become uncertain.

Real-World: Euro-area bond yields rose sharply during the ECB tightening cycle in 2022–23, implying lower bond prices. See FRED: Euro Area 10-Year Government Bond Yield.

Present Value: The Core Idea

One future payment: receiving y_{t+1} next period is worth

$$PV_t = \frac{y_{t+1}}{1+r} \quad \text{today,}$$

because investing PV_t at rate r yields exactly y_{t+1} next period.

Several future payments: discount each period separately and sum,

$$PV_t = \frac{y_{t+1}}{1+r} + \frac{y_{t+2}}{(1+r)^2} + \cdots + \frac{y_{t+n}}{(1+r)^n}.$$

Time-varying discount rates: if r changes each period,

$$PV_t(y_{t+n}) = \frac{y_{t+n}}{(1+r_{t+1})(1+r_{t+2}) \cdots (1+r_{t+n})}.$$

Intuition: a payment further in the future, or discounted at a higher rate, is worth less today. Both the *timing* and the *level* of interest rates matter.

Shares: What Does the Investor Receive?

Buy one share at the beginning of period t for price V_t .

At the end of the period, the expected payoff is

$$D_t^e + V_{t+1}^e,$$

where D_t^e is the expected dividend and V_{t+1}^e is the expected resale value of the share.

So the expected gross return on the share is

$$\frac{D_t^e + V_{t+1}^e}{V_t}.$$

Note: unlike a bond, both D_t^e and V_{t+1}^e are uncertain - they are expectations.

Required Return on Equity

The bond return r is known with certainty. The stock return is not $\rightarrow$ dividends and prices are volatile, so shares are a riskier investment.

Because investors are risk averse, they require compensation:

$$\text{expected gross return on equity} = 1 + r + \varepsilon, \quad \varepsilon \geq 0.$$

- ▶ r : time value of money (safe rate)
- ▶ ε : risk premium - extra return required to hold risky equity

Market discipline: if stocks did not offer this return, investors would switch to bonds, driving stock prices down until the condition holds.

Equity Pricing Condition

Indifference between shares and bonds requires

$$\underbrace{\frac{D_t^e + V_{t+1}^e}{V_t}}_{\text{return on equity}} = \underbrace{1 + r + \varepsilon}_{\text{required return}} .$$

Rearranging,

$$\underbrace{D_t^e}_{\text{dividend}} + \underbrace{V_{t+1}^e}_{\text{future stock price}} = \underbrace{(1 + r + \varepsilon)}_{\text{required return}} \underbrace{V_t}_{\text{stock price today}} .$$

Solving for today's stock price,

$$V_t = \frac{\underbrace{D_t^e}_{\text{dividend}} + \underbrace{V_{t+1}^e}_{\text{future stock price}}}{\underbrace{1 + r + \varepsilon}_{\text{discount factor denominator}}} .$$

Interpretation: The stock price today equals the discounted value of next period's dividend plus next period's stock price.

Interpret the Equation Carefully

$$V_t = \frac{D_t^e + V_{t+1}^e}{1 + r + \varepsilon}$$

- ▶ Numerator: expected payoff from holding the share for one period
- ▶ Denominator: discount factor based on the required return

Prices are high when expected payoffs are high and discounting (i.e. the market interest rate and the required risk premium) is low.

Forward Iteration

Apply the pricing equation one period ahead: $V_{t+1}^e = \frac{D_{t+1}^e + V_{t+2}^e}{1+r+\varepsilon}$. Insert into V_t :

$$V_t = \frac{D_t^e}{1+r+\varepsilon} + \frac{D_{t+1}^e + V_{t+2}^e}{(1+r+\varepsilon)^2}.$$

Substitute once more ($V_{t+2}^e = \frac{D_{t+2}^e + V_{t+3}^e}{1+r+\varepsilon}$):

$$V_t = \frac{D_t^e}{1+r+\varepsilon} + \frac{D_{t+1}^e}{(1+r+\varepsilon)^2} + \frac{D_{t+2}^e + V_{t+3}^e}{(1+r+\varepsilon)^3}.$$

What is happening? Each substitution replaces the resale price with the *next* period's dividend plus resale price. The resale price is pushed one period further out - and discounted by one more factor of $(1+r+\varepsilon)$.

The further out the resale price is pushed, the smaller its present value becomes. We are replacing an uncertain lump sum ("what can I sell this for?") with an ever-growing stream of dividends. In the limit, the resale price disappears entirely and only dividends remain.

General N -Step Formula

After iterating forward N times,

$$V_t = \sum_{n=0}^N \frac{D_{t+n}^e}{(1+r+\varepsilon)^{n+1}} + \frac{V_{t+N+1}^e}{(1+r+\varepsilon)^{N+1}}.$$

The last term is the discounted expected resale price very far in the future.

No-Bubbles Condition & The Fundamental Share Price

After iterating forward indefinitely, the discounted terminal resale term must vanish:

$$\lim_{N \rightarrow \infty} \frac{V_{t+N+1}^e}{(1+r+\varepsilon)^{N+1}} = 0.$$

Why is this reasonable? The stock price cannot keep growing so fast that its discounted value remains positive forever without being backed by future dividends. Otherwise, the price would reflect a bubble rather than fundamentals.

The Fundamental Share Price

The market value of the shares in a firm equals the present discounted value (PDV) of the expected future dividends paid out by the firm, where the discount rate is the market interest rate, r , plus the required risk premium, ε :

$$V_t = \sum_{n=0}^{\infty} \frac{D_{t+n}^e}{(1+r+\varepsilon)^{n+1}}.$$

Dividends and profits are closely linked; profits are the source of dividends. Hence maximizing the PDV of profits is roughly equivalent to maximizing market value.

Special Case 1: Constant Expected Dividends

Assume

$$D_{t+n}^e = D^e \quad \text{for all } n \geq 0.$$

Then

$$V_t = \sum_{n=0}^{\infty} \frac{D^e}{(1+r+\varepsilon)^{n+1}} = \frac{D^e}{r+\varepsilon}.$$

A firm paying a constant dividend forever is worth exactly the annual payout divided by the required return: higher payouts or lower required returns each raise the price proportionally.

Special Case 2: Constant Dividend Growth (Gordon Formula)

Assume dividends grow at constant rate $g < r + \varepsilon$:

$$D_{t+n}^e = (1 + g)^n D_t^e.$$

Geometric series with ratio $\frac{1+g}{1+r+\varepsilon} < 1$:

$$V_t = \frac{D_t^e}{r + \varepsilon - g}$$

The denominator $r + \varepsilon - g$ is the *net discount rate*. A firm whose dividends grow faster gets a higher valuation because each future payment is larger.

This is the Gordon Growth Model. It underlies much of applied equity analysis.

► See Derivation

Comparative Statics from the Gordon Formula

$$\underbrace{V_t}_{\text{stock price today}} = \frac{D_t^e}{r + \varepsilon - g}$$

- ▶ $D_t^e \uparrow \Rightarrow V_t \uparrow$ higher payoffs are worth more today
- ▶ $r \uparrow \Rightarrow V_t \downarrow$ future dividends are discounted more heavily
- ▶ $\varepsilon \uparrow \Rightarrow V_t \downarrow$ higher risk reduces what investors will pay
- ▶ $g \uparrow \Rightarrow V_t \uparrow$ faster dividend growth raises the entire future stream

Real-world anchors:

- ▶ ECB rate hikes 2022–23: $r \uparrow$ compressed equity valuations across the euro area.
- ▶ COVID-19 crash March 2020: $\varepsilon \uparrow$ and $D_t^e \downarrow$ simultaneously $\rightarrow$ valuations collapsed fast.

Why Stock Prices Are So Volatile

From the Gordon formula:

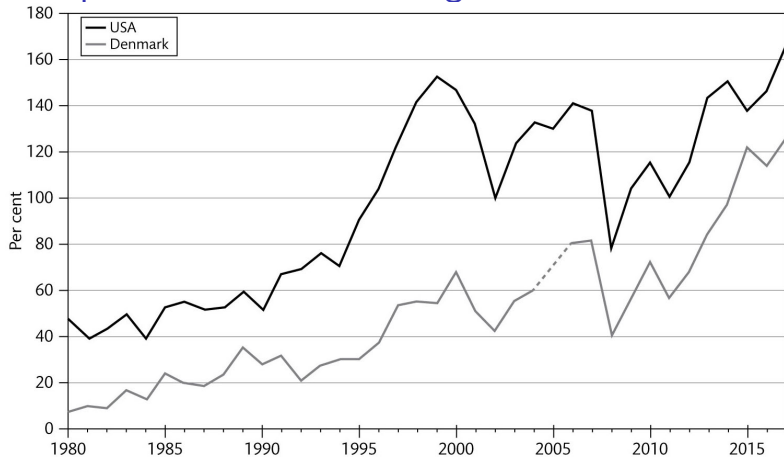
$$V_t = \frac{D_t^e}{r + \varepsilon - g}$$

Even a small change in g or $r + \varepsilon$ can produce a large swing in V_t , because the denominator is small.

Example: if $r + \varepsilon - g = 0.02$ and g rises by 0.005, the denominator falls by 25% and V_t jumps by 33%.

- ▶ Shifts in expectations about long-run growth (g) matter a lot.
- ▶ Small perceived changes in risk (ε) produce large price swings.
- ▶ This is why financial markets are inherently volatile even when fundamentals are stable.

Stock Market Capitalization as Percentage of GDP



- ▶ Stocks have become an important component of financial wealth in the USA and Denmark.
- ▶ Stock holdings can be direct and indirect (e.g. via pension funds, life insurance).

Why Stock Prices Can Lead the Business Cycle

- ▶ Asset markets react immediately to news about future profits and discount rates.
- ▶ Physical investment reacts slowly: planning, procurement, and implementation take time.
- ▶ Therefore stock prices often move *before* investment and output.

This is one reason why asset prices are useful leading indicators in macroeconomics.

Real-world example: The US stock market fell sharply in late 2022 as the Fed signalled aggressive rate hikes. Business investment in equipment slowed noticeably only in 2023 – several quarters later.

From Asset Prices to Investment: Tobin's q

- ▶ The firm cares about its market value V_t . Maximizing V_t is equivalent to maximizing the owners' expected dividends and end-of-period wealth.
- ▶ Key question: what level of investment I_t maximizes V_t ?
- ▶ To answer this, we need to link the market value of the firm to its capital stock. This is exactly what **Tobin's q** does.

Define Tobin's Average q

Normalize the acquisition price of one unit of capital to 1 (so replacement value of capital stock $= K_t$).

Let

- ▶ V_t^a : actual market value of the firm
- ▶ K_t^a : actual capital stock (at replacement cost)

Then average Tobin's q is

$$q_t \equiv \frac{V_t^a}{K_t^a} = \frac{\text{market value of the firm's capital stock}}{\text{replacement value of the firm's capital stock}}.$$

Note the direct link to stock prices: if the share price rises, V_t^a rises and q_t rises correspondingly.

An important advantage: q_t is empirically measurable since both stock market values and replacement costs of business assets can in principle be observed.

Interpretation of q_t

- ▶ $q_t > 1$: one unit of capital is worth more *inside the firm* than it costs to buy in the market.
 - ▶ Intuition: installing more capital increases firm value by more than it costs. Invest.
- ▶ $q_t < 1$: installed capital is worth less than its replacement cost.
 - ▶ Intuition: the firm would do better to shrink. Disinvest (or at least let capital depreciate).
- ▶ $q_t = 1$: market value exactly equals replacement cost. No net incentive to expand or contract.

Real-world example: During the dot-com boom (late 1990s), average q in the US reached 2–3. Investment surged (as theory predicts).

But the episode also illustrates a limit of the theory: high q reflects *market expectations* of future profits, which can be wrong. When profits failed to materialize, q collapsed and investment fell sharply.

From Average q to a Theory

The definition $q_t = V_t^a / K_t^a$ is just accounting. To turn it into a theory, we need an assumption.

Assumption (average = marginal q): For capital stocks K_t near the actual K_t^a , the firm's value is linear in K_t :

$$V_t = q_t K_t$$

This says the marginal value of one more unit of capital equals the average q_t . It is an approximation, and it is what makes the theory empirically testable.

Under this assumption, the firm can determine next period's capital stock via investment I_t :

$$K_{t+1} = K_t + I_t \quad (\text{abstracting from depreciation for now})$$

Adjustment Costs

If $q_t > 1$, why does investment not jump to infinity?

Answer: installing capital is costly. Adjustment costs include:

- ▶ Costs of installing new machinery
- ▶ Costs of training workers to use new equipment
- ▶ Costs of organizational change when the firm expands rapidly

We assume adjustment costs are *increasing and convex*:

$$c'(I_t) > 0, \quad c''(I_t) > 0$$

- ▶ $c'(I_t)$: the **marginal adjustment cost** of one more unit of investment
- ▶ $c''(I_t) > 0$: marginal adjustment costs rise with investment

A simple functional form is

$$c(I_t) = \frac{a}{2} I_t^2 \quad \Rightarrow \quad c'(I_t) = a I_t.$$

With adjustment costs, the firm's optimal condition becomes:

$$q_t = 1 + c'(I_t)$$

The Optimal Investment Rule: Intuition

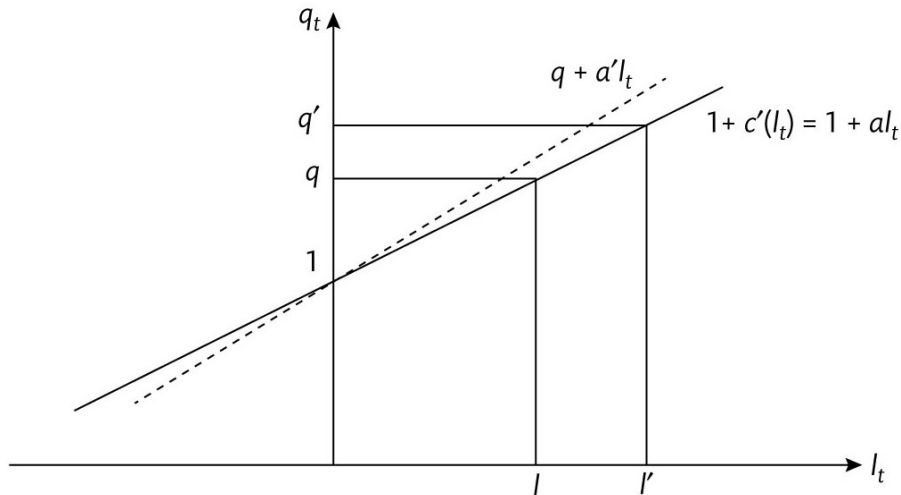
$$q_t = 1 + c'(I_t)$$

Read this as: **marginal benefit of investing = marginal cost of investing.**

- ▶ **Marginal benefit** (q_t): installing one extra unit of capital raises the firm's market value by q_t (the stock market's valuation of one unit of capital).
- ▶ **Marginal cost** ($1 + c'(I_t)$): buying the unit costs 1, plus the marginal installation cost $c'(I_t)$.

The firm keeps investing until these are equated. Higher q_t means it is optimal to push I_t further before the marginal cost catches up.

Figure 15.5: The Optimal Level of Investment



A higher q_t (shift from q to q') leads to higher investment (l to l').
A steeper cost curve (higher a) reduces the optimal investment level.

What Determines q_t ? Linking to Profits and Interest Rates

Assume real dividends are expected to stay constant at D^* from period t onwards. Then (from forward substitution of the pricing equation):

$$V_t = \frac{D^*}{r + \varepsilon}$$

Using $V_t = q_t K_t^a$:

$$q_t = \frac{D^*/K_t^a}{r + \varepsilon}$$

So:

- ▶ q_t rises with the profit rate D^*/K_t^a (better profitability raises firm value)
- ▶ q_t falls with the real interest rate r (higher discounting reduces firm value)
- ▶ q_t falls with the risk premium ε (more uncertainty reduces willingness to invest)

The Investment Function: Summary

With quadratic adjustment costs,

$$c(I_t) = \frac{a}{2} I_t^2, \quad \text{so} \quad I_t = \frac{q_t - 1}{a}.$$

If the firm value of installed capital is

$$q_t = \frac{D^*/K_t}{r + \varepsilon},$$

then substituting into the investment rule gives

$$I_t = \frac{1}{a} \left(\frac{D^*/K_t}{r + \varepsilon} - 1 \right).$$

This implies a more general investment function:

$$I = I(Y, r, \varepsilon, E, K)$$

with comparative statics

$$I(Y^+, \bar{r}, \bar{\varepsilon}, \bar{E}^+, \bar{K}).$$

Higher output Y raises expected profits and therefore q . A higher capital stock K lowers the profit rate per unit of capital and therefore reduces investment.

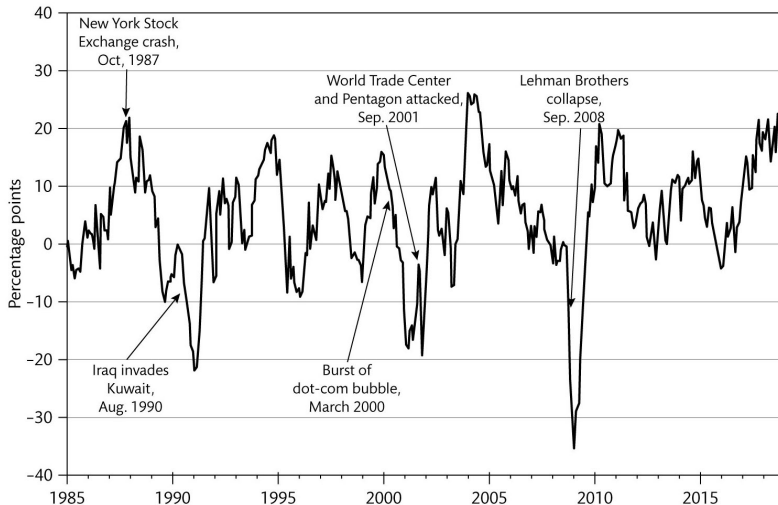
Animal Spirits and Business Confidence

Expected future dividends D^* depend not only on current profits Π_t but also on *subjective* expectations E about future demand, productivity, and prices.

We can approximate: $D^* \approx \Pi_t \cdot E$, where E is a confidence variable capturing whether investors are optimistic or pessimistic about the future.

- ▶ Most advanced economies publish *business confidence indices* that survey managers about expected future output, sales, employment, and investment.
- ▶ These indices fluctuate considerably and are affected by unanticipated political and economic events.
- ▶ Although confidence is important for investment demand, it is difficult to incorporate into theory as other than an exogenous variable.

Figure 15.7: Business Confidence in the USA



Business confidence collapses at key events: Iraqi invasion of Kuwait (1990), dot-com burst (2000), 9/11 (2001), Lehman Brothers (2008). These are precisely the episodes with investment contractions.

Figure 15.6: Tobin's q and the Investment Gap, USA 1990–2017



Tobin's q (left axis) and the investment gap – deviation of gross investment from HP trend (right axis). The two tend to co-move, but the relationship is not tight. Part of the reason: average $q \neq$ marginal q , and intangible assets are hard to measure.

Why the q -Investment Relationship Is Imperfect

The empirical relationship between measured q and investment is positive but loose. Three main reasons from the book:

1. **Average $\neq$ marginal q .** Measured $q = V/K$ is the average ratio. What should drive investment decisions is the *marginal* value of an additional unit of capital. These coincide only under the proportionality assumption.
2. **Intangible assets.** Stock prices also reflect know-how, goodwill, and brand value – assets not in the capital stock K_t . This inflates the numerator of q without reflecting investment incentives.
3. **Animal spirits.** Investment is strongly influenced by investor confidence E , which is hard to measure and not in q .

Discussion Question

A stock market boom can come from at least three sources:

1. Higher expected dividends
2. Lower real interest rates
3. Lower risk premia

In groups of 3 to 4: which of these channels is most relevant after an expansionary monetary policy shock? Which channel dominated in 2021–22?

Notebook Part: Numerical Implementation

Now we turn to the `ipynb` notebook.

- ▶ Present value and bond pricing
- ▶ Gordon growth model and comparative statics
- ▶ Forward iteration and convergence
- ▶ Tobin's q : linking stock prices to investment incentives

Summary: Key Equations from Today

$$P_t^B = \frac{1}{1+r} \quad (\text{bond price})$$

$$V_t = \frac{D_t^e + V_{t+1}^e}{1+r+\varepsilon} \quad (\text{one-period stock pricing equation})$$

$$V_t = \sum_{n=0}^{\infty} \frac{D_{t+n}^e}{(1+r+\varepsilon)^{n+1}} = \frac{D_t^e}{r+\varepsilon-g} \quad (\text{Gordon formula})$$

$$q_t \equiv \frac{V_t^a}{K_t^a} \quad (\text{Tobin's average } q)$$

$$q_t = 1 + c'(I_t) \quad (\text{optimal investment rule, derived in Lecture 2})$$

What You Should Take Away

- ▶ Asset prices are forward-looking: they reflect expected future profits and discount rates.
- ▶ Stock prices are sensitive to small changes in r , ε , and g – which explains their volatility.
- ▶ Tobin's q converts the stock market valuation of capital into an investment incentive.
- ▶ Business investment depends positively on output and confidence, negatively on interest rates and risk.
- ▶ The q -investment link is theoretically clean but empirically imperfect, mainly because of intangibles and the average vs. marginal q problem.

Preview of Lecture 2

Next lecture (remaining Chapter 15):

- ▶ Full derivation of the business investment problem with adjustment costs
- ▶ Step-by-step derivation of $q_t = 1 + c'(I_t)$ from first principles
- ▶ Quadratic example: $I_t = (q_t - 1)/a$
- ▶ Housing investment and housing q
- ▶ One Denmark-based data exercise on housing investment

Appendix

Special Case 2: Constant Dividend Growth (Gordon Formula)

Start from the fundamental price of equity:

$$V_t = \sum_{n=0}^{\infty} \frac{D_{t+n}^e}{(1+r+\varepsilon)^{n+1}}.$$

Assume dividends grow at a constant rate g :

$$D_{t+n}^e = (1+g)^n D_t^e.$$

$$V_t = \sum_{n=0}^{\infty} \frac{(1+g)^n D_t^e}{(1+r+\varepsilon)^{n+1}} = \frac{D_t^e}{1+r+\varepsilon} \sum_{n=0}^{\infty} \left(\frac{1+g}{1+r+\varepsilon} \right)^n.$$

This is a geometric series with ratio

$$x = \frac{1+g}{1+r+\varepsilon}, \quad x < 1 \text{ if } g < r + \varepsilon.$$

Using

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x},$$

$$V_t = \frac{D_t^e}{1+r+\varepsilon} \cdot \frac{1}{1 - \frac{1+g}{1+r+\varepsilon}} = \boxed{\frac{D_t^e}{r+\varepsilon-g}}.$$

How to Replicate the Denmark Figure

Turn to the `ipynb` notebook of Lecture 1.

- ▶ Open the section **Part 1: Replicate the Denmark Figure**.
- ▶ Download the two Denmark series, compute the output gap, and reproduce the chart from the slides.
- ▶ Compute the mean, standard deviation, and correlation.